

PART A [25 Marks]

This section consists of 3 questions. Answer **all** questions.

1. Find $\frac{dy}{dx}$ for the following equations:

a) $y = 3e^{2x+1}$ [3 marks]

b) $y = \sin(2 - 3x)$ [3 marks]

b) $y = \frac{x^2}{3+x}$ [5 marks]

2. Find the limit of the following, if it exists.

a) $\lim_{x \rightarrow 4} \left(\frac{x-4}{4+3x-x^2} \right)$ [3 marks]

b) $\lim_{x \rightarrow 25} \frac{x-25}{\sqrt{x}-5}$ [3 marks]

c) $\lim_{x \rightarrow \infty} \frac{\sqrt{3x^4+x}}{x^2-6}$ [3 marks]

3. Given that $f(x) = x^3 - 3x^2 - 9x + 11$. Find the stationary points of the function and determine the nature of the points.

[5 marks]

PART B [75 marks]

This section consists of 7 questions. Answer **all** questions.

1. a) Simplify $\frac{4 \cdot 3^n - 27 \cdot 3^{n-2}}{3^{n+3} - 8 \cdot 3^{n+1}}$ [4 marks]

b) Express the complex number $z = \frac{1+i}{1-2i}$ in the polar form. [6 marks]

2. Solve

a) $\log_3 3x + \log_x 3 = \log_9 3^6$ [5 marks]

b) $\frac{3}{x} \leq \frac{4}{x-1} - 1$ [6 marks]

3. Given $f(x) = \frac{kx-1}{x+2}$, $x \neq -2$. Find the value of k such that

$$f^{-1}(x+k) = -\left(\frac{3}{x} + 2\right). \quad [6 \text{ marks}]$$

4. A function f is defined by $f(x) = \ln(2x - 5)$.

a) State its domain and the range of f . [2 marks]

b) By using algebraic method, show that $f(x)$ is a one to one function. [3 marks]

c) Find the inverse of $f(x)$. [4 marks]

d) Sketch the graphs of f and f^{-1} on a single diagram. [3 marks]

END OF QUESTION PAPER

5. a) Find the values of p and q that will make the given function

$$f(x) = \begin{cases} x+2, & x < 2 \\ \frac{1}{2}px^2 - qx + 3, & 2 \leq x \leq 3 \\ 2x - p + q, & x > 3 \end{cases}$$

continuous for all real values of x .

[6 marks]

b) Evaluate $\lim_{x \rightarrow 0} \left(\frac{2}{x} - \frac{2}{x\sqrt{x+1}} \right)$

[5 marks]

- 6 (a) Given the parametric equations $x = 3 \cos 2t$ and $y = 12 \sin t$.

Show that $\frac{dy}{dx} = -\operatorname{cosec} t$.

[4 marks]

- (b) A function f is defined by

$$f(x) = \begin{cases} -1 - \frac{1}{x^2}, & x < 1, \\ 3 - x^2, & x \geq 1 \end{cases}$$

determine whether the function f is differentiable at $x = 1$.

[5 marks]

- (c) If $y^3 = \ln(x^3 y^2)$ for $x > 0, y > 0$, then find $\frac{dy}{dx}$ when $y = 1$. [7 marks]

7. Water is poured into an inverted cone of depth 10 cm and radius 5 cm at a rate of $8 \text{ cm}^3 \text{ s}^{-1}$. Find the rate at which the depth of the water in the cone is increasing when the depth is 4 cm.

[9 marks]

